The Simson–Lu Theorem

— Collinearity After n−2 Iterations of the Pedal Transformation of a Cyclic n-gon

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Timeline

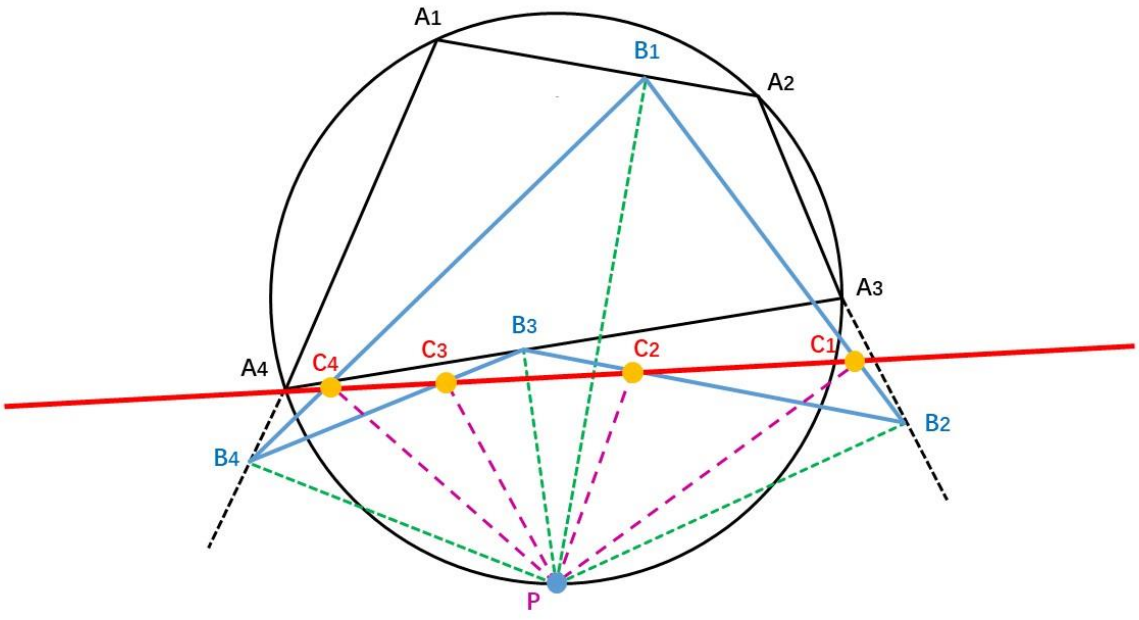
2002 (age 17, during high school): The author reports discovering the theorem.

2007: The author reports public publication in Popular Science magazine.

2022: Video materials were posted on Bilibili.

2023: A complex-number proof was discussed by another contributor on Mathematics Stack Exchange.

September 2026: With AI assistance, the author organized four proof approaches: elementary geometry, analytic geometry, complex numbers, and higher mathematics.



Abstract

This paper studies the following recursive pedal construction. Let $A_1A_2\ldots A_n$ be n points on a circle, and let P be another point on the same circle. In each round, orthogonally project P onto the supporting lines through consecutive vertices; then repeat the operation using the lines through consecutive points of the resulting pedal polygon. The paper claims that after n−2 iterations the n resulting points are collinear, and that after n−1 iterations all n points coincide. For n=3, the statement reduces to the classical Simson–Wallace theorem.

Four proof frameworks are presented: (1) elementary Euclidean geometry using inscribed angles, pedal circles, and a Miquel-type geometric recursion; (2) analytic geometry and induction, expressing the feet in coordinates; (3) a complex-number approach using normalization to the unit circle and a product formula; and (4) an operator-theoretic approach interpreting the iteration as a sliding-product operator on a periodic sequence, with shift operators, circulant matrices, and discrete Fourier analysis used to describe the claimed one-dimensional degeneration.

Keywords: Simson theorem; pedal polygon; pedal transformation; cyclic polygon; mathematical induction; complex method; linear operator; discrete Fourier transform.

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1. Introduction

The classical Simson–Wallace theorem states that if P lies on the circumcircle of triangle ABC, then the three perpendicular feet from P to the lines containing the sides of the triangle are collinear. This is a basic result in pedal geometry. For a general polygon, one pedal transformation does not usually produce collinear points; however, the resulting pedal polygon can itself be transformed repeatedly. This paper concerns the finite-step degeneration claimed for this iterative pedal process.

The central claim is that, for a cyclic n-gon, repeated pedal transformations with a fixed point P on the circumcircle produce a line degeneration at round n−2 and a point degeneration at round n−1:

n-gon → round n−2 → n collinear points → round n−1 → n coincident points.

This recursive pattern is presented as a generalization of the classical case n=3: a triangle requires one round, a quadrilateral two rounds, a pentagon three rounds, and a general n-gon n−2 rounds.

Historically, polygonal pedal constructions are not new. Steggall's 1895 work discussed extensions of the Simson line for polygons; in 1961, Zylbertrest published "A Generalisation of Simson's Theorem," describing the polygon formed by orthogonal projections onto the sides as the first pedal. A 2023 public discussion also presented a cyclic-polygon problem with a recursion of the same form and an algebraic proof. Accordingly, this paper's contribution is positioned as a systematic presentation of the construction and four mathematical proof frameworks, rather than a claim of historical priority without further documentary verification.

2. Definitions, Notation, and Theorem Statement

2.1. Geometric Construction

Let Γ be a nondegenerate circle, let $A_1,A_2,\ldots,A_n$ be points on Γ in cyclic order, and let $P \in \Gamma$. Indices are taken modulo n, so $A_{n+1}=A_1$. Define the points in round 0 by

A_i^(0)=A_i, i=1,2,…,n.

For r≥0, once the points A_i^(r) have been constructed, define A_i^(r+1) as the orthogonal projection of P onto the line A_i^(r)A_{i+1}^(r). These are supporting lines, so the foot need not lie on the corresponding segment.

For convenience in the figures, successive rounds are denoted by successive letters: B_i, C_i, and D_i correspond to A_i^(1), A_i^(2), and A_i^(3), respectively. In general, the points in round r in the figures represent A_i^(r) as defined in the text.

A_i^(r+1)=Proj_{A_i^(r)A_{i+1}^(r)}(P).

2.2. Nondegeneracy Assumptions

The main theorem is stated for the usual nondegenerate case: the construction lines are well-defined and the initial n-gon has no repeated vertices. If points coincide or an edge degenerates at an intermediate step, the result may be interpreted by continuity. The paper claims that its central algebraic formula also yields the final degeneration in such cases.

2.3. Main Theorem

Theorem 1 (the Simson–Lu theorem, as named in this paper). Let $A_1A_2\ldots A_n$ be a cyclic n-gon on Γ, and let $P \in \Gamma$. Apply the pedal transformation repeatedly according to the rule above. Then

$A_1^{(n-2)}, A_2^{(n-2)}, \ldots, A_n^{(n-2)}$ are collinear.

Moreover, the stronger conclusion is claimed:

$A_1^{(n-1)}=A_2^{(n-1)}=\ldots=A_n^{(n-1)}$.

Thus, the collinearity statement is presented as a direct consequence of the stronger claim that all points coincide one round later.

3. Elementary Geometric Proof: Inscribed Angles and Miquel-Type Recursion

This section seeks to use as little coordinate machinery as possible, organizing the argument around perpendicularity, inscribed angles, right-angle circles, a Miquel-type chain of circles, and uniqueness of a perpendicular or parallel line. For general n, a fully expanded angle chase would have many branches; the paper therefore compresses the geometric recursion into a local pedal-recursion lemma and outlines its intended rationale.

3.1. A Basic Pedal-Circle Lemma

Lemma 1. Let P be fixed, let U and V be two points on a line, and let H be the foot of the perpendicular from P to UV. Then the configuration P,H,U,V has a right-angle-circle structure involving PU or PV as a corresponding chord. In particular, any line through H perpendicular to UV coincides with PH.

Proof. Since $PH \perp UV$, $\angle PHU = \angle PHV = 90°$. Thus the right triangles with hypotenuses PU and PV are constrained by the same foot-of-perpendicular configuration. Repeating this construction for adjacent sides, a sequence of quadrilaterals can be shown cyclic by the criterion that opposite angles sum to 180°. This is the local basis intended for the subsequent Miquel-chain argument.

3.2. Triangle Case: The Classical Simson Theorem

Let P lie on the circumcircle of triangle ABC, and let D, E, F be the feet from P to BC, CA, and AB, respectively. Using ordinary (unsigned) length ratios, the right triangles PBD, PCD, PCE, PAE, PAF, and PBF give

$$|BD|/|DC| = PB\,|\cos\angle PBC| \,/\, (PC\,|\cos\angle PCB|).$$

$$|CE|/|EA| = PC\,|\cos\angle PCA| \,/\, (PA\,|\cos\angle PAC|).$$

$$|AF|/|FB| = PA\,|\cos\angle PAB| \,/\, (PB\,|\cos\angle PBA|).$$

Multiplying gives

$$(BD/DC)(CE/EA)(AF/FB) = -1,$$

Since P, A, B, and C are concyclic, the inscribed-angle theorem gives

$$\angle PBC = \angle PAC, \quad \angle PCB = \angle PAB, \quad \angle PCA = \angle PBA.$$

The three groups of cosine factors therefore cancel. The product of the ordinary length ratios is 1; restoring the signs of directed segments gives the product −1 required by Menelaus' theorem. Hence D, E, and F are collinear.

$$\angle DEF = \angle DPE = \angle DPC = \angle DFC \pmod{180°}.$$

Thus $\angle DEF = \angle DFC$, so DE and DF are the same line, and D, E, F are collinear. This is the classical Simson–Wallace theorem.

3.3. Geometric Recursion Lemma

Lemma 2 (iterated-pedal degeneration lemma). Suppose all vertices of an initial cyclic polygon lie on the same circle as P. Apply the pedal transformation repeatedly, using P as the fixed projection point. If the adjacent edges in round r are connected by one Miquel circle chain, then the directed-angle relations of the local quadrilaterals in round r+1 are determined entirely by r+2 consecutive vertices of the original polygon. When the consecutive window covers all n initial vertices, all final feet are claimed to be the same point.

Proof sketch. For any consecutive window $A_i, A_{i+1}, \ldots, A_{i+r+1}$, drop perpendiculars from P to the successive edges. Each new foot introduces a right-angle circle; adjacent circles share a foot and P and are intended to combine, by Miquel's theorem, into one circle chain. The inscribed-angle theorem makes the directed angles of adjacent construction lines depend only on the endpoints of the window and P. Advancing the window by one vertex preserves the intermediate construction and adds one endpoint. When the window has length n, each starting index covers the same set of n original points, so the final feet are asserted to have identical geometric definitions and therefore coincide.

3.4. Deriving Collinearity in Round n−2 from Coincidence in Round n−1

Let S be the common point in round n−1. By the definition of the pedal transformation, S is the foot from P to every line in round n−2:

$A_i^{(n-2)} A_{i+1}^{(n-2)}$.

Each of these lines is therefore perpendicular to PS. Through a point in the plane there is a unique line perpendicular to PS, so all the edges in round n−2 must be the same line L. Hence

$A_1^{(n-2)},\ldots,A_n^{(n-2)} \in L.$

This proves the theorem, provided the claimed coincidence in round n−1 has been established.

Note. The purely geometric section emphasizes the geometric structure and intended recursion. Since a full angle chase for general n is not supplied, the paper relies on the analytic, complex-number, and operator sections for its claimed computational verification.

4. Analytic-Geometric Proof: Coordinate Formula and Mathematical Induction

4.1. Coordinate Normalization

Translations, rotations, and similarity scalings preserve collinearity, perpendicularity, and the structure of the pedal construction. We may therefore take the circumcircle to be the unit circle

$$x^2+y^2=1$$

and set P=(1,0). Let two consecutive points on the circle be

$$A=(\cos\alpha, \sin\alpha), \quad B=(\cos\beta, \sin\beta).$$

Set

$$m=(\alpha+\beta)/2, \quad d=(\alpha-\beta)/2.$$

4.2. Equation of Chord AB

Using product-to-sum identities, the line AB has equation

$$x\cos m + y\sin m = \cos d.$$

Since A and B satisfy this equation, it is the line containing chord AB. Its unit normal vector is

$$n=(\cos m, \sin m).$$

4.3. The Foot from P to AB

The signed distance from P=(1,0) to the line AB is cos m−cos d. Thus the foot F is

$$F=P-(\cos m-\cos d)n.$$

Using $\cos m-\cos d=-2\sin(\alpha/2)\sin(\beta/2)$, we obtain

$$F=(1,0)+2\sin(\alpha/2)\sin(\beta/2)(\cos m, \sin m).$$

For any point X=(cos θ, sin θ) on the circle, define the two-dimensional quantity

$$Q(\theta)=((1-\cos\theta)/2, -\sin\theta/2)=(\sin^2(\theta/2), -\sin(\theta/2)\cos(\theta/2)).$$

For A=(cosα,sinα) and B=(cosβ,sinβ), write Q(α)=(a,b) and Q(β)=(c,d). From the foot-coordinate formula in Section 4.3, the foot F=(x,y) satisfies

$$x=1-2(ac-bd), \quad y=-2(ad+bc).$$

Consequently, Q(F)=((1−x)/2,−y/2)=(ac−bd,ad+bc). Define the binary operation on R^2 by

$$(a,b)\odot(c,d)=(ac-bd, ad+bc).$$

Lemma 3. Under the coordinate normalization above, let $A_i^{(r)}$ denote the i-th point in round r and set $Q_i^{(r)}=Q(A_i^{(r)})$. Then

$$Q_i^{(r)}=Q_i\odot Q_{i+1}\odot\ldots\odot Q_{i+r}.$$

Proof. For r=1, the foot-coordinate formula directly gives

$$Q_i^{(1)}=Q_i \odot Q_{i+1}.$$

Assume that for some $r\geq 1$,

$$(u\odot v)\odot w=u\odot(v\odot w),$$

and that the foot formula has the same form for adjacent windows. Then

$$Q_i^{(r+1)}=Q_i^{(r)} \odot Q_{i+r+1}=Q_i \odot Q_{i+1} \odot \dots \odot Q_{i+r+1}.$$

The claimed formula follows by induction.

4.5. Periodicity and Final Coincidence

Because there are n initial points, indices are taken modulo n. When $r=n-1$, every $Q_i^{(n-1)}$ contains the same n factors $Q_1,\dots,Q_n$, only in cyclically shifted order. Since $\odot$ is commutative and associative,

$$Q_1^{(n-1)}=Q_2^{(n-1)}=\dots=Q_n^{(n-1)}.$$

Thus all n points in round $n-1$ coincide. By the uniqueness-of-perpendicular-line argument in Section 3.4, the n points in round $n-2$ are collinear.

4.6. Advantages of the Analytic-Geometric Approach

It does not use complex-number notation; its basic tools are coordinates, dot products, trigonometric identities, and induction.

The key recursive idea is that round r+1 adds just one new adjacent-vertex factor.

The final collinearity is not isolated; it is derived from the claimed complete coincidence in round $n-1$.

4.7. Illustrations

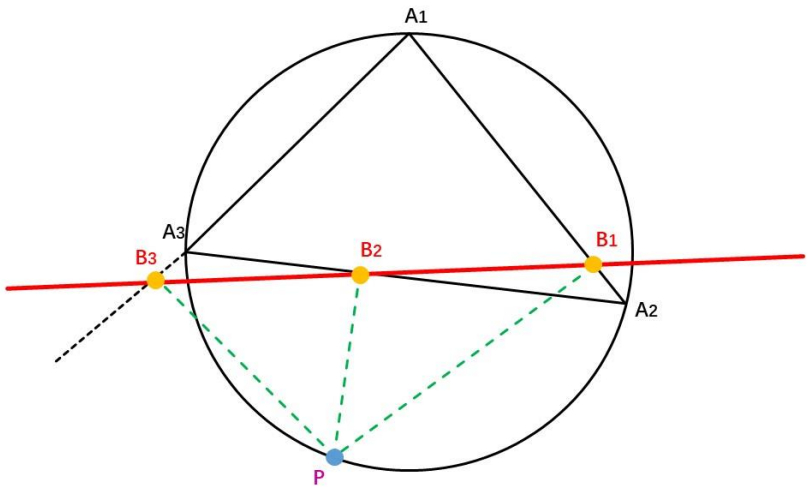


Figure 1. Triangle case: after one pedal transformation, the three feet are collinear.

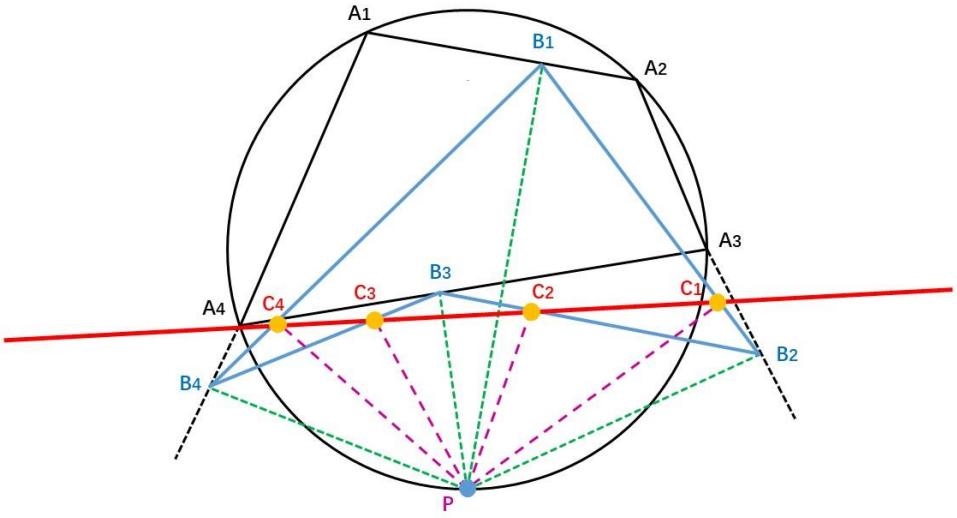


Figure 2. Quadrilateral case: after two pedal transformations, the four points are collinear.

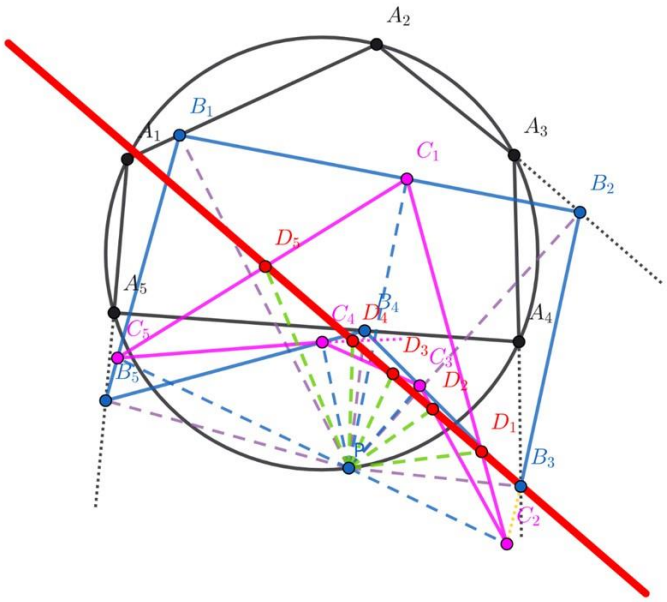


Figure 3. Pentagon case: after three pedal transformations, the five points are collinear.

5. Complex-Number Proof: Unit Circle and Product Invariant

5.1. Normalization in the Complex Plane

Apply a similarity transformation to normalize the circumcircle to the unit circle |z|=1 and set P=1. For any complex number z, define

$$[z]=(1-z)/2.$$

This notation encodes the position of z relative to P=1 as a complex factor.

5.2. Projection Lemma

Lemma 4. Let |u|=|v|=1, and let W be the foot from 1 to the line uv. Then

$$[W]=[u][v].$$

Proof. Set q=[u]=(1−u)/2 and s=[v]=(1−v)/2. Write u=e^{iα} and v=e^{iβ}; then

$$q=-i\, e^{i\alpha/2} \sin(\alpha/2), \quad s=-i\, e^{i\beta/2} \sin(\beta/2).$$

Let h=qs. The half-angle expressions are said to show that h lies on the line qs and that h−0 is perpendicular to s−q. Thus h is the foot from the origin O to the line qs. Now consider the similarity

$$T_M(z)=1-2Mz.$$

It maps O to 1 and the line qs to T_M(q)T_M(s), preserving perpendicularity. For M=1, $T_1(q)=u$ and $T_1(s)=v$, so $T_1(h)=W$. Hence

$$[W]=(1-W)/2=qs=[u][v].$$

This proves the lemma.

5.3. Consecutive Product Formula

Let Y_i^(r) be the i-th point in round r, with indices taken modulo n. By Lemma 4, the first round satisfies

$$[Y_i^{(1)}]=[Y_i][Y_{i+1}].$$

Assume that

$$[Y_i^{(r)}]=[Y_i][Y_{i+1}]\dots[Y_{i+r}].$$

Let

$$M=[Y_{i+1}][Y_{i+2}]\dots[Y_{i+r}].$$

Then

$$Y_i^{(r)}=1-2[Y_i]M,$$

$$Y_{i+1}^{(r)}=1-2[Y_{i+r+1}]M.$$

The points $Y_i^{(r)}$ and $Y_{i+1}^{(r)}$ are the images of $[Y_i]$ and $[Y_{i+r+1}]$ under the similarity T_M. Applying the projection lemma again gives

$$[Y_i^{(r+1)}]=[Y_i][Y_{i+1}]\dots[Y_{i+r+1}].$$

By induction,

$$[Y_i^{(r)}]=\prod_{j=0}^{r}[Y_{i+j}],\quad r\geq 1.$$

5.4. Coincidence After n−1 Rounds and Collinearity After n−2 Rounds

Set $r=n-1$. Since indices are taken modulo n, each product contains each of $[Y_0],\dots,[Y_{n-1}]$ exactly once. Therefore

$$[Y_0^{(n-1)}]=\dots=[Y_{n-1}^{(n-1)}].$$

The map $z\mapsto[z]=(1-z)/2$ is a one-to-one affine transformation, so

$$Y_0^{(n-1)}=\dots=Y_{n-1}^{(n-1)}=S.$$

S is the foot from P to every edge in round n−2. All those edges are therefore perpendicular to PS and must lie on the same line. This proves the stated conclusion, provided the preceding product recurrence is valid.

6. Higher-Mathematical Approach: Similarity Operators, Cyclic Shifts, and Fourier Structure

6.1. Treating the Pedal Transformation as an Operator

Rather than viewing each perpendicular construction in isolation, this section treats it as an operator acting on a sequence of points. Define the projection operator

$$\square_P(A_i,A_{i+1})=\mathrm{Proj}_{A_iA_{i+1}}(P).$$

With the unit-circle normalization and $P=1$, introduce the affine similarity operator

$$T_M(z)=1-2Mz.$$

When $M\neq 0$, T_M is a similarity of the complex plane, combining rotation, scaling, and translation. In particular, it preserves collinearity and perpendicularity.

6.2. Conjugating the Operator: From Geometric Projection to Multiplication

Let u and v lie on the unit circle, and set $q=[u]$, $s=[v]$. By the projection lemma in Section 5, the foot from the origin to qs is qs. Under T_M, the foot from 1 to $T_M(q)T_M(s)$ is

$$T_M(qs)=1-2Mqs.$$

Thus, if $B(z)=[z]$ denotes the normalized coordinate, the pedal operator is claimed to be conjugate in B-coordinates to the product of two adjacent factors. For the full n-point sequence, this yields the sliding product

$$B(Y_i^{(r)}) = B(Y_i)B(Y_{i+1})\dots B(Y_{i+r}).$$

This is presented as the key bridge to operator theory: repeated orthogonal projections are encoded by a fixed window-product operator.

6.3. Cyclic Shift Operator

Let the vector be

$$b=(b_0,b_1,\dots,b_{n-1})^T,$$

and define the cyclic shift operator E by

$$E(b_0,b_1,\dots,b_{n-1})^T=(b_1,b_2,\dots,b_{n-1},b_0)^T.$$

Clearly,

$$E^n=I.$$

If $b_i\neq 0$ in a generic configuration, local complex logarithms $\ell_i=\log b_i$ can be chosen. The product formula for round r then becomes additive:

$$\ell_i^{(r)}=\ell_i+\ell_{i+1}+\dots+\ell_{i+r}.$$

Hence

$$\ell^{(r)}=S_r\,\ell^{(0)},\quad S_r=I+E+E^2+\dots+E^r.$$

6.4. Rank of the Cyclic Operator at the Final Round

When $r=n-1$,

$$S_{n-1}=I+E+\dots+E^{n-1}.$$

For every vector b, each component of $S_{n-1}b$ equals $b_0+\dots+b_{n-1}$. Therefore

$$\mathrm{Im}(S_{n-1})=\mathrm{span}\{(1,1,\dots,1)\}.$$

The image of S_{n-1} is one-dimensional. In other words, after $n-1$ window accumulations, all positions lose their differences and only a common value remains. This is the linear-algebraic picture offered for the claimed coincidence of all final points.

6.5. Discrete Fourier Decomposition

The cyclic shift E can be diagonalized by the discrete Fourier matrix F. Let $\omega=e^{2\pi i/n}$; then

$$E=F^{-1}\mathrm{diag}(1,\omega,\omega^2,\dots,\omega^{n-1})F.$$

Consequently,

$$S_r=F^{-1}\mathrm{diag}(\Sigma_{j=0}^{r}\omega^{kj})_{k=0}^{n-1}F.$$

In particular, when $r=n-1$, for $k\neq 0$,

$$\Sigma_{j=0}^{n-1}\omega^{kj}=0,$$

whereas the sum is n for $k=0$. Thus

$$S_{n-1}=F^{-1}\mathrm{diag}(n,0,\dots,0)F.$$

This says that all $n-1$ nonconstant discrete Fourier modes are eliminated, leaving only a one-dimensional constant mode. The paper then infers coincidence in round $n-1$ from the injectivity of the product coordinate, and collinearity in round $n-2$ from the foot construction.

Technical note. The Fourier analysis is intended to explain and characterize the rank-one degeneration. The geometric identity itself is supposed to follow from the projection lemma and consecutive-product formula in

Section 5. If some [Y_i]=0, logarithms are unnecessary: the product formula would make all final products zero, leading to the same conclusion.

7. Unified Mechanism of the Four Approaches and Low-Order Examples

7.1. Correspondence Between the Four Approaches

7.2. Triangle

$$n=3 \Rightarrow n-2=1.$$

Thus one pedal transformation gives three collinear points, which is the classical Simson–Wallace theorem.

7.3. Quadrilateral

$$n=4 \Rightarrow n-2=2.$$

The first round gives four pedal points; the second gives four collinear points; in the third round all four points coincide.

7.4. Pentagon

$$n=5 \Rightarrow n-2=3.$$

The first, second, and third rounds successively increase the length of the consecutive window. In the fourth round all final feet coincide; therefore the five points in the third round are collinear.

7.5. General n-gon

Stable structure of the n-gon iteration: $0 \to 1 \to \ldots \to n-2 \to n-1$.

Round $n-2$ is the line degeneration and round $n-1$ is the point degeneration. This hierarchy—first reducing to one dimension, then to zero dimensions—is presented as the central structure revealed by the four approaches.

7.6. Original Construction Figure

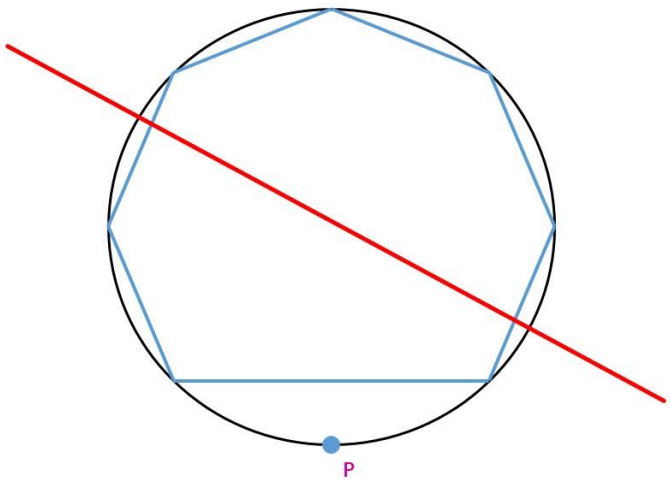


Figure 4. General n-gon case: after n−2 pedal transformations, the n points are collinear.

8. Degeneracies, Scope, and Scholarly Boundaries

8.1. Supporting Lines Rather Than Finite Segments

Throughout this paper, "dropping a perpendicular to an edge" means dropping it to the supporting line of the edge, in the standard analytic-geometric sense; the foot is not required to lie inside the segment. This convention gives a uniform definition for any cyclic n-gon and any point P on its circumcircle.

8.2. Coincidence of P with a Vertex

If P coincides with an initial vertex, the corresponding $[Y_i]$ is zero. The product formula is claimed to remain valid, and every product in round $n-1$ contains this zero factor, so all final points coincide.

8.3. Repeated Vertices and Completely Degenerate Polygons

If the initial "n-gon" has repeated vertices, it is not a nondegenerate polygon in the usual sense. The main theorem is intended for nondegenerate cyclic polygons; completely degenerate cases are outside its scope.

8.4. Relation to Earlier Work

Steggall's 1895 paper discussed extensions of the Simson line to polygons; Zylbertrest's 1961 paper called the polygon formed by projecting a point onto the sides the first pedal and studied its properties. A 2023 Mathematics Stack Exchange question describes a recursive construction of the same form as the one considered here and gives an algebraic proof.

9. Conclusion

This paper presents a unified study of iterative pedal transformations of a cyclic n-gon with respect to a fixed point P on the circumcircle. Its central claim is that after $n-2$ pedal transformations the n points are collinear, and after one further transformation, in round $n-1$, all n points coincide.

The four approaches describe the same claimed structure at different mathematical levels: elementary geometry invokes right-angle circles and a Miquel chain; analytic geometry expresses each round as a coordinate recursion and uses induction; the complex method turns the pedal transformation into a consecutive product; and the operator-theoretic approach interprets the product as a window accumulation under cyclic shifts and uses Fourier decomposition to explain the remaining constant mode.

The theorem may therefore be described not only as a collinearity statement but as a finite-step degeneration:

$$n\text{-gon} \rightarrow \text{one-dimensional degeneration (a line) at order } n-2 \rightarrow \text{zero-dimensional degeneration (a point) at order } n-1.$$

This formulation is intended as a unified starting point for further study of iterative pedal polygons, discrete geometric transformations, and periodic operators.

References

[1] Alison, J. "The so-called Simson line." Proceedings of the Edinburgh Mathematical Society, 1884, 3: 77–93. DOI: 10.1017/S0013091500037354.

[2] Steggall. "On the envelope of the Simson line of a polygon." Proceedings of the Edinburgh Mathematical Society, 1895, 14: 122–126. DOI: 10.1017/S0013091500031874.

[3] Zylbertrest, S. "A Generalisation of Simson's Theorem." The Mathematical Gazette, 1961, 45(351): 30–37. DOI: 10.2307/3614768.

[4] Ferrarello, D.; Mammana, M. F.; Pennisi, M. “Pedal Polygons.” International Journal of Mathematical Education in Science and Technology, 2013. (Cited for background on general pedal polygons.)

[5] Tsukerman, E. “On Polygons Admitting a Simson Line as Discrete Analogs of Parabolas.” arXiv:1201.0305, 2012.

[6] Zeta. “A generalization of Simson line to cyclic polygons.” Mathematics Stack Exchange, 2023.

[7] Blue. Answer to “A generalization of Simson line to cyclic polygons.” Mathematics Stack Exchange, 2023. (The answer gives a product-type algebraic proof for a recursion of the same form.)

Appendix A. Four Construction Figures

The following four figures are from the author’s earlier construction materials and are intended to correspond to the recursive pedal process for triangles, quadrilaterals, pentagons, and general n-gons.

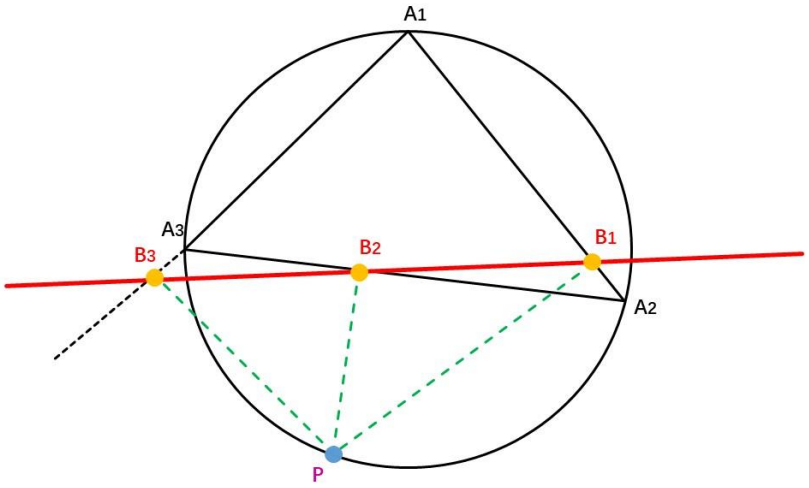


Figure 1. Triangle case: the three feet are collinear after one pedal transformation.

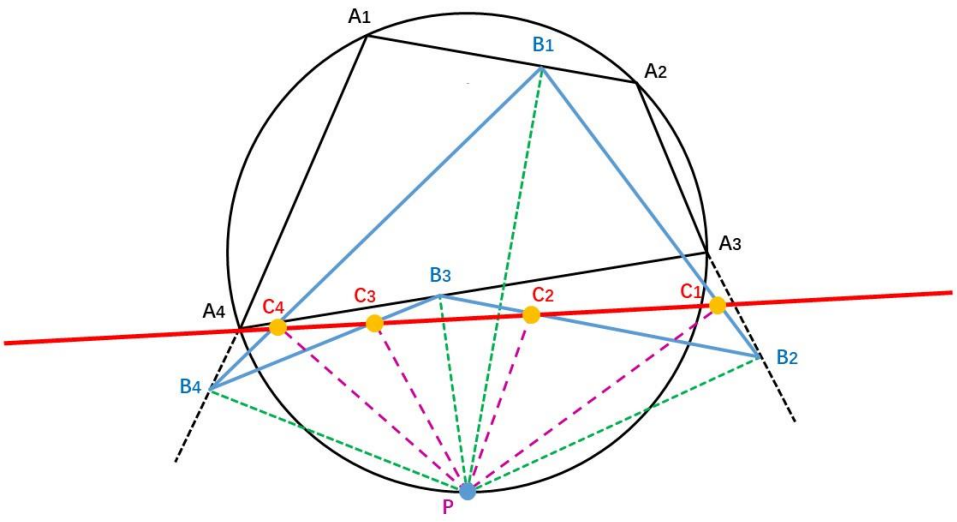


Figure 2. Quadrilateral case: the four points are collinear after two pedal transformations.

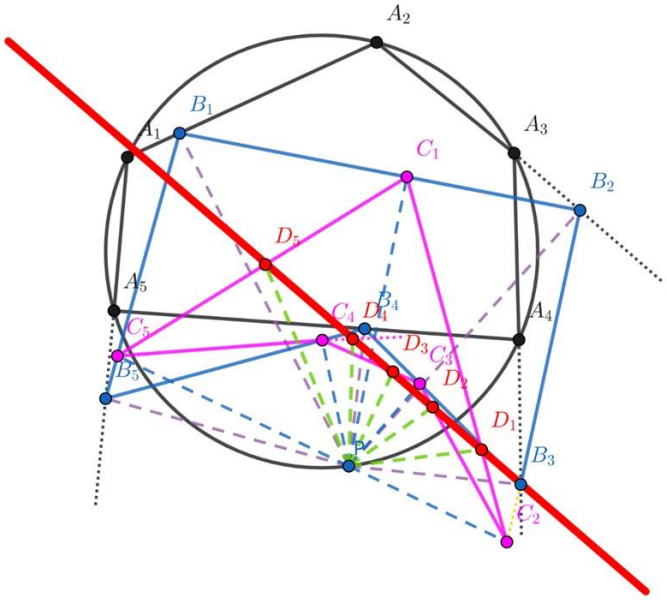


Figure 3. Pentagon case: the five points are collinear after three pedal transformations.

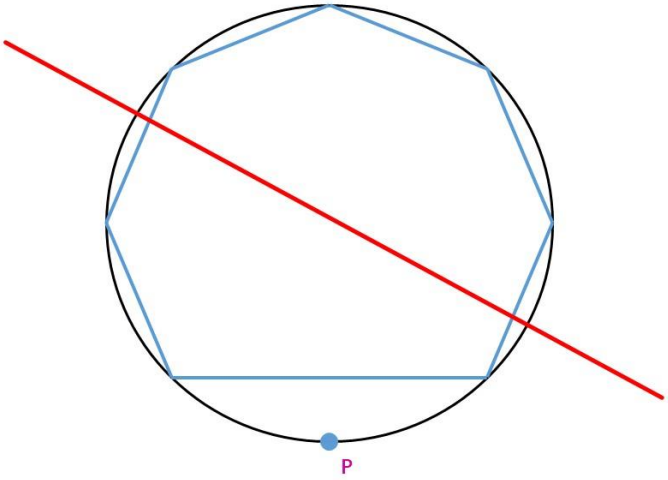


Figure 4. General n-gon case: the n points are collinear after n−2 pedal transformations.

Appendix B. Notation

Table 1

Approach	Core tools	Key intermediate result	Final conclusion
Elementary geometry	Inscribed angles, right-angle circles, Miquel chain	All final feet coincide	Edges in the penultimate round coincide
Analytic geometry	Coordinates, trigonometric identities, induction	Consecutive-window recursion	n−1 轮重合
Complex method	Unit circle, similarity, product	[Y_i^(r)]=∏[Y_{i+j}]	n−2 轮共线
Operator-theoretic approach	Shift operators, circulant matrices, DFT	S_{n−1} 秩为 1	Final degeneration to a point

Table 2

Symbol	Meaning
Γ	Initial circumcircle
P	Fixed point on the circle and projection point in each round
A_i	i-th initial point on the circle
A_i^(r)	i-th pedal point in round r
[z]	Complex normalization (1−z)/2
E	n-dimensional cyclic shift operator
S_r	Window accumulation operator I+E+…+E^r
T_M	Affine similarity T_M(z)=1−2Mz
⊙	解析几何部分定义的二维乘法 (a,b)⊙(c,d)=(ac−bd,ad+bc)